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Investigation of the urinary metabolic variations and the application in bladder cancer biomarker discovery

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Abbreviations: BMI: body mass index; AUC: area under the curve; LC-HRMS: liquid chromatography- high resolution mass spectrometry; CSF: cerebrospinal fluid; QC: quality control; PCA: principal component analysis; OPLS-DA: orthogonal partial least squares discriminant analysis partial least squares discriminate analysis; ROC: receiver operator characteristic curve; VIP: variable importance plots; CV: coefficient of variation; BCa: bladder cancer

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Novelty and Impact: In this study, we conducted a comprehensive characterization of the urine metabolome of 203 healthy subjects and 53 patients with bladder cancer. Inter-individual analysis indicated that the urine metabolome was relatively stable and could be used for biomarker discovery. Further analysis showed that age and sex were two important factors associated with urine metabolome variation. A pilot urinary metabolome characterization in bladder cancer by age- and sex-matched urine samples was able to distinguish cases with bladder cancer from controls, and high-grade bladder cancer from low-grade. The novel biomarker panels were discovered to have predictive ability with an AUC of 0.956 for detecting the presence of bladder cancer and 0.937 for high- and low-grade bladder cancer distinction. External validation indicated an accuracy of 90% for detecting bladder cancer and 75% for detecting high-grade bladder cancer.

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Abstract

Urine metabolomics have been used to identify biomarkers for clinical diseases. However, inter-individual variations and effect factors need to be further evaluated. In the present study, we explored the urine metabolome in a cohort of 203 health adults, 6 patients with benign bladder lesions, and 53 patients with bladder cancer (BCa) using liquid chromatography coupled with high resolution mass spectrometry. Inter-individual analysis of both healthy controls and bladder cancer patients showed that the urine metabolome was relatively stable. Further analysis indicated that sex and age affect inter-individual variations of urine metabolome. Metabolic pathways such as tryptophan metabolism, the citrate cycle, and pantothenate and CoA biosynthesis were found to be related to sex and age. To eliminate age and sex interference, additional BCa urine metabolomic biomarkers were explored using age and sex-matched urine samples (Test group: 44 health adults vs. 33 patients with BCa). Metabolic profiling of urine could significantly differentiate the cases with cancer from the controls and high-grade from low-grade BCa. A metabolite panel consisting of trans-2-dodecenoylcarnitine, serinyl-valine, feruloyl-2-hydroxyputrescine, and 3-hydroxynonanoyl carnitine were discovered to have good predictive ability for BCa with an area under the curve (AUC) of 0.956 (cross validation: AUC=0.924). A panel of indolylacryloylglycine, N_2-galacturonyl-L-lysine, and aspartyl-glutamate was used to establish a robust model for high and low-grade BCa distinction with AUC of 0.937 (cross validation: AUC=0.891).External sample(26 control vs. 20 BCa) validation verified the acceptable accuracy of these models for BCa detection.This study showed that urinary metabolomics is a useful strategy for differential analysis and biomarker discovery.

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Introduction

Urine examination has long been used in scientific research because of its diagnostic value in the clinic. Compared to other body fluids, urine is collected non invasively and can be obtained in large amounts. Urine metabolomics has become a useful strategy to identify biomarkers for diseases[1-3]. To obtain useful diagnostic information, as many small-molecule metabolites as possible must be quantified and compared with normal samples. However, the definition of the normal sample is a challenging task due to the large individual variations among humans. It has been established that the sex, age, body mass index (BMI)[3, 4], and even the humans'chronotype or sample collection time [5, 6] could contribute to the metabolic composition of urine.Therefore, perturbations in these elements must be considered during the design of experiments. Moreover, understanding metabolites' variation with physiological factors (such as sex and age) in healthy humans will provide a basis for comparative research as well as insight into metabolic mechanisms of a healthy body.

A few researchers have been investigating urine metabolite variations in healthy humans. Due to the high polar property of urine metabolites, previous research used NMR technology to explore urinary profiling variations correlating with changes in physical conditions. NMR data from rat urine by Bollard et al. suggested an influence of various physiological effects including age, sex, diet, species, and hormonal status on urine metabolic profiling[4]. In humans, several studies using a small cohort of subjects (60 or 24 volunteers, respectively) indicated these metabolites had variations correlating with sex and age, such as TCA cycle intermediates, creatine concentrations, and fatty acid oxidation[7, 8]. In 2015, Etienne A. collected 183 subjects' urine samples and studied the correlation of urinary metabolites with age, BMI, and sex using the LC-MS platform. A total of 108 metabolites were found to be correlated with age, sex, or BMI.

For the magnification effects of metabolomics on inter-individual variations compared to proteomics, and the multiple physiological factors that influence the level of metabolites, it is commonly thought that metabolites are not sufficiently stable compared with proteins for clinical biomarker applications[9, 10]. However, no researchers have performed a comprehensive inter-individual variation metabolomics analysis to date. Only a few small cohort studies have explored the effects of physiological factors on the metabolome, which might induce bias in the statistical differential analysis. Until 2015, when the large-scale study by Etienne A mentioned above was published, age was continuously divided into 5groups, but the sample size for each group is so small in that study that the statistical significance of age variations needs to be further validated.

Bladder cancer (BCa) is a common urinary tract cancer worldwide. Recent reports in 2016 showed hat more than 21% death rate was estimated in the U.S. and 40% death rate happened in Taiwan[11]. BCas are categorized as low-grade and high-grade. Nowadays, cystoscopy and cytology are standard methods for BCa detection[12]. Although cytology is a noninvasive method and has a high specificity, its sensitivity is not satisfactory, particularly for low-grade tumors[13]. There is a great need to further improve the diagnosis of malignant bladder tumors. Most of the recent research efforts have been directed at discovery of BCa biomarkers using omics technologies. Many genes or proteins have been reported to have potential value for BCa diagnosis, such as thrombospondin-1, uromodulin, coronin-1A, and apolipoprotein A4 and A1 for BCa detection [14, 15], and S100A8 and S100A4 for high-grade and low-grade BCa distinction[16].

Urine is in direct contacted with bladder epithelia cells that may give rise to BCa, and metabolites

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released from bladder cancer cells may be enriched in urine samples. Therefore, urine metabolomics is a promising approach for BCa detection and marker discovery. In 2014, Xing Jin et al. used HPLC-MS to perform urine metabolomics for 138 BCa patients and 121 controls. They identified 12 differential metabolites involved in glycolysis and beta-oxidation that contributed to BCa distinction with an AUC of 0.937[17]. Bryan M. et al. analyzed urine metabolomics of 66 BCa positive samples using UHPLC-MS/MS and GC-MS. They found a metabolite panel of palmitoylsphingomyelin, lactate, adenosine, and succinate had the strongest predictive power for differentiating urine samples from patients with BCa cancer from those without, with an AUC of 0.81[18]. Recently, in 2017, Chi-hung et al. studied the urine metabolomics of 87 BCa patients and 65 hernia controls using UPLC-TOF-MS. They found a predictive model based on six metabolites for BCa detection with an accuracy 84.76%, but only one marker, imidazoleacetic, was identified[11]. Thus, there have been multiple studies on BCa detection and marker discovery based on the urine metabolome. However, few researchers have focused on the distinction between high- and low-grade BCa diagnosis. The identified putative markers may be affected by the variable, dynamic, and diverse nature of the urine metabolome, and future urine metabolic profiling is necessary to discover new urinary markers.

In the present work, we collected urine samples from 203 Chinese volunteers, with a balanced sample size based on sex and each age group (Table 1) to significantly explore the influence of age and sex on urine metabolome. Furthermore, using urine metabolomics, BCa biomarkers were explored based on age- and sex-matched healthy subjects and patients with BCa (Table 1). Potential urine biomarkers were discovered to have a predictive ability with an AUC of 0.956 for distinction between patients with BCa and controls and 0.937 for distinction between high- and low-grade BCa. Further external validation supported the acceptable accuracy of BCa prediction through the use of the potential biomarkers. This study will provide new insights into BCa diagnosis using urine metabolome.

Materials and methods

Urine Collection

This study was approved by the Institutional Review Board of the Institute of Basic Medical Sciences, Chinese Academy of Medical Sciences. All human subjects provided informed consent before participating in this study. The first morning urine (midstream) was collected at around07:00 to 09:00 a.m. on an empty stomach from a cohort of 203 clinically healthy human adults, 6 patients with benign bladder lesions,and53 patients with BCa at Beijing Tiantan Hospital.Their detailed demographics and disease pathologyare shown in Table 1 and Table S1.More than90% of the patients were diagnosed as non muscle invasive stage, thus we focused on metabolome comparison between different grades (low vs. high).

Urine Sample Preparation

Acetonitrile (200 µl) was added to each urine sample (200 µl), then the mixture was vortexed for 30 s and centrifuged at 14,000 × g for 10 min. The supernatant was dried under vacuum and then reconstituted with 200 µl of 2% acetonitrile. Urinary metabolites were further separated from larger molecules using 10 kDa molecular weight cut-off ultra centrifugation filters (Millipore Amicon Ultra, MA) before transferred to the autosamplers. The quality control (QC) sample was a pooled urine sample prepared by mixing aliquots of fifty representative samples across different groups to be analyzed and therefore globally representative of the whole sample set. The QC samples were injected

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every ten samples throughout the analytical run to provide a set of data from which method stability and repeatability can be assessed.

LC-MS Analysis

Ultra-performance LC-MS analyses of urine samples were conducted using a Waters ACQUITY H-class LC system coupled with a LTQ-Orbitrap mass spectrometer (Thermo Fisher Scientific, MA. USA). Urinary metabolites were separated with a 29 min gradient on a Waters HSS C18 column (3.0 × 100 mm, 1.7 μm) at a flow rate of 0.3 ml/min. The mobile phase A was 0.1% formic acid in H_2O and the mobile phase B was acetonitrile. The gradient was set as follows: 0–2 min, 2% solvent B; 2–5 min, 2–55% solvent B; 5–15 min, 55–100% solvent B; 15–20 min, 100% solvent B; 20–20.1 min, 100–2% solvent B; 20.1–29 min, 2% solvent B. The column temperature was set at 50℃. All samples were full-scan analyzed from 100 to 1000 m/z at a resolution of 60 K. The automatic gain control (AGC) target was 1 × 106 and the maximum injection time (IT) was 100 ms.Subsequently, differential metabolite identification was performed by UPLC targeted-MS/MS analyses of the QC sample.It was acquired at a resolution of 15 K with an AGC target of 5×10^5, maximum IT of 50 ms, and isolation window of 3 m/z. Collision energy was optimized as 20, 40, 60, or 80 for each target with higher-energy collisional dissociation (HCD) fragmentation. The injection order of urine samples with three technical replicates was randomized to reduce any experimental bias.

Data processing

Raw data files were processed by the Progenesis QI (Waters, Milford, MA, USA) software based on a previously published identification strategy[19, 20].The detailed workflow for QI data processing and metabolites identification is given in supplementary material 1.Further data pre-processing including missing value estimation, log transformation and Pareto scaling were carried out to make features more comparable using MetaAnalyst 3.0 (http://www.metaboanalyst.ca). Variables missed in 50% or greater of the samples, were removed from further statistical analysis. Non-parametric tests (Wilcoxon rank-sum test) were used to evaluate the significance of variables. False discovery rate (FDR) correction was used to estimate the chance of false positives and correct for multiple hypothesis testing.The adjusted *p*-value (FDR) cutoff was set as 0.05. Pattern recognition analysis (principal component analysis, PCA; orthogonal partial least squares discriminant analysis, OPLS-DA) was carried out using SIMCA 14.0 (Umetrics, Sweden) software. The selected differential variables must have met the following three conditions: 1) adjusted P-value <0.05; 2) fold change between two groups >1.5; 3) the VIP value obtained from OPLS-DA was above 1.5. Exploratory ROC analysis and external biomarker validation were carried out using the "Biomarker discovery" module on the MetaAnalyst 3.0 platform.

Results

The workflow of this study is shown in Fig. 1. A total of 262 subjects were enrolled, with 203 volunteers that were clinically healthy, 6 with benign bladder lesions, and 53 patients that were clinically diagnosed with BCa. Metabolite variations associated with age and sex were explored based on metabolic profiling analysis of 203 healthy subjects. BCa biomarkers were identified based on profiling analysis of 44 age- and sex-matched healthy subjects and 33 BCa patients.Furthermore, a pilot differential analysis was performed to discriminate urinary metabolic profiling between high-grade and low-grade BCa.The potential biomarkers were further externally validated using an independent batch of patients with BCa (8 low-grade and 12 high-grade) and control samples

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(including 6 with benign lesions and 20 samples from healthy individuals).

Quality control

This large cohort of samples was analyzed randomly, which took almost a week. QC is of great importance in large-scale metabolomics studies to ensure stable system performance and to limit experimental bias. A QC standard was prepared as a pooled mixture of aliquots from representative urine samples in each group. The QC sample was injected 5 times before and frequently throughout the analytical run to monitor instrument stability. Overall, 32 injections were performed during the whole analysis. Metabolomics technical reproducibility was assessed by analyzing the QC sample variations with time. The first 6 injections showed a large variation due to the unbalanced status of the instrument, which was followed by a stable condition with only a small variation (< ± 2 SD) (Fig. S1a)[21]. The tight clustering of QC samples (Fig. S1b) further demonstrated the quality of the QC data. These results provided some assurances that the platform had essential repeatability and stability throughout the analytical run.

Inter-individual variations in the urinary metabolome

Inter-individual variations Overall, 2642 features could be quantified in the normal urinary metabolome and used for individual variation comparisons. The median of the inter-individual CV (coefficient of variation) was 0.941 (Fig. 2a). The inter-individual variation in the female urinary metabolome was similar to that in the male urinary, with 0.891/0.890, respectively. The variation in the female or male profiles was smaller than the whole, suggesting that sex might contribute to urine metabolome variation. For age variation, the CV in the older urinary metabolome was a little higher than that in the younger urine, and the CV in each age group was smaller than that in the whole, suggesting that age contributed to urine metabolome variation. The detailed information is shown in Table 2 and Fig. S2.

In BCa urinary metabolome, a total of 3945 features could be quantified. The inter-individual CV was a little smaller than that in normal urine, with a median of 0.833 (Fig. 2b). More quantified variations and smaller inter-individual CV than those in the normal urine samples partly resulted from the small BCa sample size. Similar to the normal urine metabolome, the CV in females and males was smaller than the whole, suggesting that sex contributes to BCa urine metabolome variation. The detailed information is shown in Table 2 and Fig. S2.

According to the following differential analysis, a total of 98 differential (age and sex-related)metabolites were identified in normal urine subjects. The median of the inter-individual CV for these metabolites was 0.923 (Fig. S2c). For cases of BCa, overall 106 differential metabolites were identified. The median of the inter-individual CV was 1.103 (Fig. S2d). The trend of the inter-individual variations for the identified differential metabolites was similar to the whole metabolome. The detailed information is provided in Table S2 and Fig. S2.

Correlating individual variation and metabolite abundance Peak areas were used to evaluate metabolite abundance in this study. The log2 transformed for each metabolite was plotted against the corresponding inter-individual variation. No correlations between inter-individual CVs and metabolite abundance in normal urine were found. Similar to the results from normal urine metabolites, the individual variation did not correlate with metabolite abundance in urine from patients with BCa (Fig. S2). These results show that the inter-individual variations are not correlated with metabolome abundance in urine. For these identified metabolites, we arrived at the same conclusion (Fig. S2).

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Sex variations

According to our results, sex and age are important factors that influence the urine metabolome, which has been reported previously[7, 22].We further explored the detailed metabolites associated with sex and age. First, unsupervised PCA was performed to explore the tendency of metabolic profiling variations between males and females. The resulting score plot is shown in Fig. S3a. It shows that the male group could be separated from the female group, with most male samples clustering to the right of the PCA plot, although some male samples overlapped with the female group. Class separation could also be obviously observed for males and females using an OPLS-DA model (Fig. 2c). One hundred permutation validations of the model produced R^2= 0.92, Q^2= 0.67, Q^2-incept = −0.331. The difference between R^2 and Q^2 was small and the incept of the Q^2 line crossed the zero axis, which is an indication of a no-overfit model. Analysis of OPLS-DA variable importance plots (VIP), combined with statistical test analysis ($p < 0.05$), indicated several metabolites responsible for the sex difference (Table S3).

Around one-third of the differential metabolites showed higher levels in females, including 3-hydroxyanthranilic acid, N-acetyltryptophan, and uric acid. Pathway power analysis indicated that tryptophan metabolism, the citrate cycle, pantothenate, and CoA biosynthesis contributed significantly to the sex variation (Fig. S3b and S3c). In males, the steroid hormone biosynthesis pathway was activated, and showed significant statistical differences; however, this contributed less to sex differences.

To test if these significant metabolites had any predictive ability for sex discrimination, multivariate ROC curve based exploratory analysis was carried out (http://www.metaboanalyst.ca/faces/Secure/upload/RocUploadView.xhtml). A metabolites panel comprising pantothenic acid, 5-hydroxyindoleacetic acid, 3-hydroxyanthranilic acid, oxoglutaric acid, and 3-sulfodeoxycholic acid was used to construct the most robust predictive model for sex discrimination. On the training set, this model classified male vs. female with an AUC ROC of 0.967. The cross-validated AUC ROC was 0.958, a relatively modest difference from 0.967, indicating no over-fitting (Fig. S3d). The sensitivity and specificity of the model was above 0.90 (Table S4). By comparing the results without biomarker selection (AUC ROC of 0.70 for all features), the established model demonstrated significantly increased discriminatory power and prediction accuracy.

Age variation

PCA was performed to explore the tendency of metabolic profiling variations with age. Three groups, including the young (aged 20–30), the middle-aged (aged 40–50), and the old (aged > 60), were compared. The score plot showed a relatively low correlation of the metabolic profiling with age without sex classification, with an AUC of 0.698 for the old, 0.752 for the young, and 0.668 for the middle-aged (Fig. S4a and S4b). It is predicted that a different metabolism status for males and females caused a bias in the discrimination. Therefore, we further examined metabolic profiling variations with age for males and females, respectively. The results showed a better discrimination degree of these three age groups for males and females (Male: AUC (aged 20–30): 0.808; AUC (aged 40–50): 0.818; AUC (aged > 60): 0.897; Female: AUC (aged 20–30): 0.831; AUC (aged 40–50): 0.720; AUC (aged > 60): 0.742 [Fig. S4c and S4d]). The above results indicate that the different sexes showed different metabolic statuses.

To discover differential metabolites with age，OPLS-DA was used to distinguish metabolic profiling of the young samples from the middle-aged and the middle-aged from the old. An integrated

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SUS-plot obtained from the two OPLS-DA models was used to evaluate the correlation of metabolites with age. Score plots showed a clear distinction of different age groups for both females and males (Fig. 2d and Fig. S5). One hundred permutation tests showed no over-fitting of the models. Variables located on the upper left and lower right in the SUS-plot showed significantly linear changes with age, and the bigger the absolute value of X and Y, the stronger the correlation with age (Fig. S5). Moreover, an ANOVA test was performed on metabolite changes with age and only variables with $p < 0.05$ were considered to be significant. Finally, a total of 25 metabolites were identified as differential markers with age in males and 28 changed significantly with age in females (Table S5).

Based on the above results, age contributes to metabolite variations in both males and females. The functional pathways of these differential metabolites were further analyzed. The relative intensity change trend is shown in Fig. 2e and 2f. In males, metabolites involved in amino acid metabolism, such as arginine and ornithine, were up-regulated with aging, and metabolites involved in arachidonic acid metabolism were down-regulated with aging. In females, metabolites involved in arachidonic acid metabolism and proline metabolism were up-regulated with aging, while carbohydrate metabolism and tyrosine metabolism were down-regulated with aging. These results indicate that metabolism status variation with aging is different in males and females, which provided evidence for the necessity of subject-matching during experiment design.

BCa biomarker discovery

Since age and sex are important factors affecting metabolism, for BCa biomarker discovery, 44 age and sex-matched control samples out of the 203were selected to eliminate interference from age and sex. In addition, most of the BCa patients and controls were around 60 years of age. Hence, differential metabolites among patients aged 20–30, 40–50 and > 60 theoretically have no correlation with the differential metabolites in BCa, and thus the differential metabolites identified for patients with BCa could be considered to be disease-related.Pathological analysis classified 18 patients (low-grade BCa) and 15 patients (high-grade BCa). The details of the patients and control subjects are shown in Table 1. All of the study subjects were xanthoderm.

First, PCA was performed to explore the tendency of metabolic profiling variations between control and BCa subjects (Fig. S6a) and suggested apparent discrimination. Then, an OPLS-DA model was used for differential metabolite selection (Fig. 3a). Overall, 48 significantly differential metabolites were identified (Table S6). Their relative intensity was plotted as a heatmap in Fig. 3b, which shows that metabolites involved in amino acid metabolism and fatty acid oxidation are up-regulated in BCa patients, while metabolites involved in phenylalanine metabolism and xenobiotics (e.g., caffeine) metabolism were down-regulated.

To evaluate the diagnostic accuracy of these differential metabolites for BCa, a predictive model for patient classification was constructed using each identified metabolite, respectively. Overall, 33 metabolites had a potentially useful diagnostic value with the AUC above 0.7, and 11 metabolites had a good diagnostic value with the AUC above 0.8 (Table S7)[23]. It has been reported that combined biomarkers are more valuable for diagnosing disease progression than just one biomarker [17]. Multivariate ROC curve-based exploratory analysis was carried out (http://www.metaboanalyst.ca/faces/Secure/upload/RocUploadView.xhtml) using these differential metabolites and a predictive model based on the metabolite panel consisting of trans-2-dodecenoylcarnitine, serinyl-valine, feruloyl-2-hydroxyputrescine, and 3-hydroxynonanoyl

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carnitine showed the best predictive ability with a ROC area of 0.956 for the testing dataset and 0.924 for 10-fold cross-validation (Table S8).Further external validation was performed to validate the prediction accuracy of this model. The model could correctly predict 18 out of 20 BCa patients and 24 out of 26 non-BCa (healthy and benign lesions, Fig. 3c), with an accuracy of 90% for BCa prediction.

According to the pathologic evaluation of the tumor, BCa was categorized as high grade (15 samples) and low grade (18 samples). Herein, using the similar multiple statistical methods as above, the metabolic profiling differentiation was explored between high- and low-grade BCa urinary samples. The PCA plot showed a large number of overlapping samples in the two groups (Fig. S6b). Furthermore, ROC curves based on metabolic profiling were plotted to quantify the discrimination degree, which showed acceptable classification accuracy with an AUC (high-grade) 0.876 and an AUC (low-grade) 0.883 (Fig. S6c). The OPLS-DA model was further used for differential metabolites selection (Fig. 3d). Based on statistical test analysis ($p < 0.05$) and VIP value (VIP > 1.5), 58 significantly differential metabolites were identified (Table S9)**.** The relative intensity was plotted as a heatmap in Fig. 3e,which showed that metabolites involved in retinol metabolism, histidine metabolism, tryptophan metabolism, and arachidonic acid metabolism were up-regulated in high grade BCa patients, while metabolites involved in fatty acid oxidation were down-regulated. Overall, 40 metabolites had a clinically potential diagnosis value with the AUC above 0.7, and only 6 metabolites had a good diagnostic value with an AUC above 0.8 (Table S10). Using a logistic regression algorithm, a metabolite panel consisting of indolylacryloylglycine, N_2-galacturonyl-L-lysine, and aspartyl-glutamate was used to establish a robust model for high and low-grade BCa distinction. The ROC was 0.937 for the testing dataset and 0.891 for 10-fold cross-validation (Table S11).Further external validation was performed to validate the prediction accuracy of this model. The model could correctly classify 9 out of 12 high-grade BCa patients' (Fig. 3f) urine samples, showing an accuracy of 75% for high-grade BCa prediction.

Discussion

In this study, we conducted a comprehensive characterization of the urine metabolome of 203 healthy subjects, 6 patients with benign lesions and 53 patients with BCa. Individual variations were comprehensively evaluated and related factors were analyzed. Metabolic pathways that correlated with age and sex were identified. Furthermore, we identified a urinary metabolome characterization of BCa, which was able to distinguish BCa from controls and high-grade from low-grade BCa. Our study showed that urinary metabolomics is an effective strategy for differential analysis and biomarker discovery and can provide new insights into BCa diagnosis using urine metabolome.

Urine metabolomevariation

It is widely accepted that urine proteome variations are relatively stable and could be used for biomarker applications [25]. Recently, WenchuanLeng et al., explored the individual variations of 2000 commonly detected urine proteins in 497 proteome samples, which ranged from 1.01 to 1.19[26]. In our work, we evaluated the inter-individual CVs of urine metabolites in 203 normal subjects and 33 BCa patients. The median CV for 203 normal urine metabolome was 0.941 and for the 33 BCa urine metabolomes it was 0.833. These results indicate that urine metabolites are stable enough to be used as potential biomarker candidates. Moreover, sex and age are two important factors that contribute to urine metabolome variation, which must be considered during biomarker study design.

Metabolome variation with sex and age

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According to inter-individual variation analysis, age and sex are important factors that affect the urinary metabolome. Understanding the detailed differential metabolites associated with age and sex can provide a basis for comparative research as well as insight into the metabolic mechanisms of a healthy body.

Sex variation

Previous studies have found higher concentrations of testosterone and an accumulation of cortexolone in males, which is the reason for higher testosterone glucuronide[27].Our study showed consistent results that steroid hormone biosynthesis was activated in males, with higher levels of cortexolone, tetrahydrodeoxycorticosterone (THDOC), cortisol, tetrahydrodeoxycortisol, and testosterone glucuronide. THDOC is a neuroactive steroid that increases in biofluid concentration during stress. It has both positive and negative modulatory effects on GABA-activated currents[28]. A higher level of cortexolone and tetrahydrodeoxycorticosterone may indicate higher stress in males. The second cluster of metabolites with concentrations higher in males was acyl carnitines. A higher concentration of acyl-carnitines in males has recently been reported for urine and serum[22, 29], which indicates a higher activity of energy production, fatty acid metabolism, and intermediary metabolism in tissues[30]. Tryptophan metabolism, the citrate cycle, and pantothenate and CoA biosynthesis were up-regulated in females. Tryptophan is involved in a variety of processes, including the production of biogenic amines such as serotonin, melatonin, and tryptamine[31]. In mammals, it contributes to the synthesis of nicotinamide adenine dinucleotide (NAD+), a coenzyme important for energy metabolism (such as the citrate cycle). A previous study suggested that tryptophan metabolism might play a role in regulating lifespan in humans[32]. It is possible a higher level of tryptophan metabolism is one of the contributors to a longer lifespan in females. A higher level of citrate metabolism in urine from females is reported to be controlled by estrogen, which might be responsible for the underlying regulatory mechanism of metabolic pH in the mitochondria and cytoplasm and consequently citrate transport through the mitochondrial membrane[8].

Age variation

The extent of sexual dimorphism in the metabolome has recently been shown, stressing the need for sex stratification [29]. In males, metabolites involved in fatty acid metabolism, amino acid oxidation, and GSH depletion show an increased level with age. A higher level of GSH deletion suggests a decreased metabolic activity/decreased ability of free radical scavenging with age in males[33]. Amino acid oxidation will cause some proteins to lose their function and to be hydrolyzed more easily by proteases, which is the key element of aging[34]. Metabolites involved in lipid oxidation, protein, amino acid biosynthesis, and 4-hydroxyandrostenedione metabolism decreased with age in males. A decreased lipid oxidation process indicates a weakened function of mitochondria with age[33]. Accumulating evidence suggests that alterations in amino acid and protein synthesis are critical factors in the progression of ageing[35]. In females, amino acid degradation and oxidation metabolites and acyl carnitine (glutarylcarnitine) are up-regulated with age. The increased activity of amino acid degradation and oxidation with age occurred in both males and females, so these may be considered to be metabolic pathway changes associated with age in the population as a whole. The observed higher levels of glutarylcarnitine with advanced age in females might indicate that the aging processes counteract oxidative damage [36].

In addition to age and sex variation, human circadian rhythms are also important factors affecting

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urine metabolome[5], which may partly contribute to metabolome variation.

BCa biomarker discovery

Recent metabolomics studies have proven valuable in identifying urinary biomarkers for BCa diagnosis [37, 38], but obtaining an accurate and sensitive noninvasive diagnosis biomarker of human BCa remains a challenging problem. In the present study, a pattern based on 4 urinary metabolites was found to have high accuracy for BCa distinction, and a pattern based on 3 urinary metabolites, indolylacryloylglycine, N_2-galacturonyl-L-lysine, and aspartyl-glutamate, was found to have high accuracy for BCa grading distinction.

In a comparison of the healthy controls and the BCa cases in the urinary metabolic profiling, we identified a total of 48 metabolites that could significantly differentiate cases with BCa from the healthy controls. Our results suggest significant changes in pathways related to amino acids (e.g., phenylalanine, glycine, glutamate), metabolism, and energy metabolism (e.g., fatty acid oxidation) in patients with BCa. This is consistent with the profile of tumor metabolism, which needs to acquire energy from circulation for rapid ATP generation and maintenance of appropriate cellular redox status[39]. In previous BCa research, it has been suggested that metabolites involved in glycolysis, fatty acid beta-oxidation (decanoylcarnitine, octenoylcarnitine)[17], amino acid metabolism (valine, leucine, glycine), and lipid metabolism (acylcarnitines)[18]contributed to the distinction between the BC and control groups. In the present study, we found that several acylcarnitines (decanoylcarnitine, decenoylcarnitine, hydroxynonanoylcarnitine, and hydroxybutyrylcarnitine) were increased in BCa patients. Carnitine is essential for the entry of fatty acid into the mitochondria for oxidation.Our findings are consistent with previous reports. These results suggest that fatty acid oxidation might be an important factor in determining cancer status.A higher expression level of carnitine palmitoyltransferase 1A (CPT1A), a key enzyme in fatty acid oxidation, may contribute to an increase in acylcarnitines[17, 40].Moreover, xenobiotic metabolites, including aromatic compounds, hydroxylated metabolic derivatives, and caffeine metabolites (methylxanthine), were found to be altered in BCa patients, which pinpoints aberrations in xenobiotic metabolism. According to previous reports, variations of aromatic compounds and hydroxylated metabolic derivatives could be mainly attributed to defects in the genes regulating detoxification processes and methylation as an additional contributing factor in regulating the expression of xenobiotic metabolizing enzymes[41, 42].However, the mechanism of methylxanthine metabolism disorders in BCa needs to be explored further.

To address the shortcomings of urine cytology and cystoscopy for screening and grading of urinary BCa[43], we applied the metabolomics approach as a surrogate tactic for accurate BCa grading. Specifically, the differentiation of high-grade and low-grade BCa has not been explored using a urine-derived metabolomics approach. Serum-based NMR metabolomics indicated that glutamine, histidine, and valine could distinguish high-grade from low-grade, and these metabolites can be used for early prediction of BCa[44]. Pathways of fatty acid oxidation and amino acid metabolism were found to be disturbed to meet the needs of cancer cells that are undergoing a series of compensatory metabolic adaptations and altering their energy supply[45].Fatty acid metabolites (pimelylcarnitine, propionylcarnitine, cinnamylcinnamate, hydroxynonanoylcarnitine, and keto-decanoylcarnitin) were significantly up-regulated in the low-grade compared to the control group. However, in the high-grade group, only pimelylcarnitine and keto-decanoylcarnitin were higher than in the control group, and the other three metabolites had no significant differences compared with the control group. These results

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suggested different mechanisms of different types of fatty acid metabolism in high- and low-grade BCa.

In the present study, the pathways of retinol metabolism, histidine metabolism, tryptophan metabolism, and arachidonic acid metabolism were altered in high-grade BCa. Histidine metabolism is the source of histamine, which is involved in inflammation-associated transitional cell carcinoma in urinary bladders[46, 47]. It has been demonstrated that histidine metabolism in bladder cancer tissue specimens is elevated[41], and serum histidine may be a promising and early diagnostic gauge to distinguish between low-grade and high-grade BCa. These conclusions partially validated our results. Retinol metabolism is reported to be involved in bladder cancer progression[48]. Researchers have shown that retinols have a potential protective effect on bladder cancer risk, and serum retinols levels are significantly reduced in patients with bladder cancer when compared with controls[49, 50].Our results indicate the differential retinol metabolism between high and low grade BCa, which is another potential marker for BCa grading distinctions. Moreover, tryptophan metabolism was found to be up-regulated in high-grade BCa, indicating an associated increased absorption of tryptophan in patients with high-grade bladder cancer[51]..

Conclusion

In conclusion, our study conducted a comprehensive characterization of urine metabolome and demonstrated that urine metabolites were stable enough that they could be applied for biomarker discovery. However, the influence of physiological factors must be considered during study design, especially for studies with a limited sample size. Our pilot study of BCa urine metabolic profiling could distinguish the BCa group from the control group and between patients with high-grade low-grade BCa. Panels of metabolites were discovered to have potential value for BCa diagnosis and BCa grading distinction. In addition, because of hematuria, since many important signals might come from plasma instead of BCa, the metabolomics characteristics of BCa need to be compared with control hematuria samples in the future. Therefore, our data will benefit the application of urine metabolome to disease biomarker discovery.

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Conflict of interests The authors declare no conflict of interests.

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Figure legends

Fig. 1 The workflow of this study.

Fig. 2 Analysis of metabolome inter-individual variations and related factors (sex and age).**a. b.** Inter-individual variations of urine metabolome in normal and BCa subjects.**c.** Score plot of OPLS-DA model between females and males. **d.** Score plot of OPLS-DA based on urine profiling of males aged 20–30 and 40–50. **e.** Relative intensity of differential metabolites at different ages in females. **f.** Relative intensity of differential metabolites at different ages in males.

Fig. 3 Analysis of metabolic profiling in BCa.**a.** Score plot of OPLS-DA model based on metabolome between BCa and control. **b.** Relative intensity of differential metabolites in BCa. **c.**External prediction accuracy of BCa prediction model established by a metabolite panel of trans-2-dodecenoylcarnitine, serinyl-valine, feruloyl-2-hydroxyputrescine, and 3-hydroxynonanoyl carnitine.**d.** Score plot of OPLS-DA of urinary metabolic profiling of high- and low-grade BCa. **e.** Relative intensity of differential metabolites in high- and low-grade BCa. **f.**External prediction accuracy of high-grade BCa prediction model established by the metabolite panel of indolylacryloylglycine, N2-galacturonyl-L-lysine, and aspartyl-glutamate.

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Table 1 Demographics of healthy subjects and BCa patients

	Male	Female	Total
Healthy subjects			
Aged 20-30	22	20	42
Aged 30-40	16	18	34
Aged 40-50	12	18	30
Aged 50-60	22	25	47
Aged >60	26	24	50
BCa patients (discovery group)			
BCa all (Aged 63.8±13.2)	**21**	**12**	**33**
Low-grade (**Age**d 63.2±16.4)	11	7	18
High-grade (**Age**d 63.2±8.8)	10	5	15
Health control (Aged 65.0±11.3)	**28**	**16**	**44**
BCa patients (validation group)			
BCa all (Aged 61.8±14.3)	**14**	**6**	**20**
Low-grade (**Age**d 55.3±16.7)	6	2	8
High-grade (**Age**d 66.1±11.2)	8	4	12
Control all (Aged 54.3±14.1)	**17**	**9**	**26**
Health (**Age**d 61.7±11.9)	12	8	20
Benign (**Age**d 53.0±19.7)	5	1	6

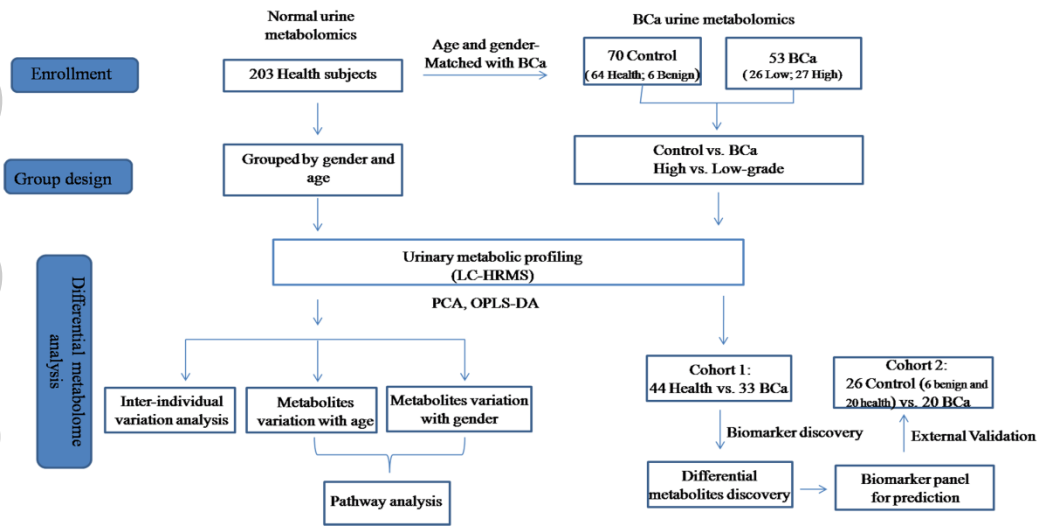
Table 2 The inter-individual variations of the urinary metabolome in control and BCa subjects

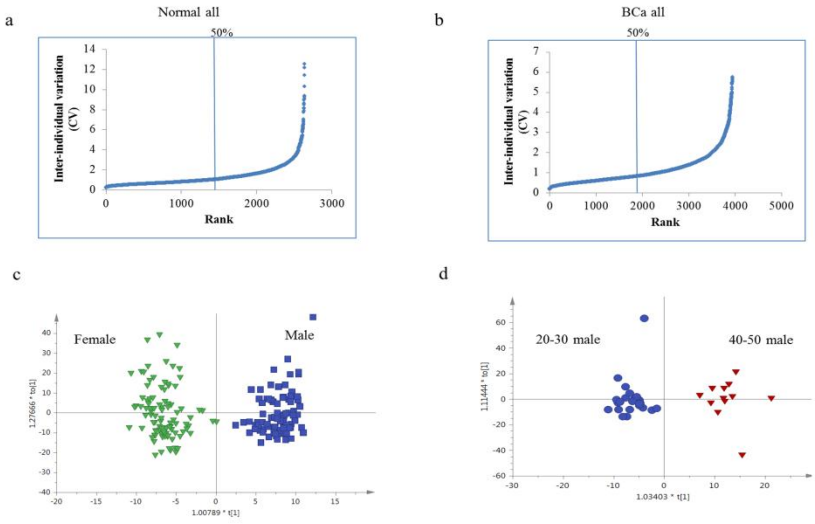
	Median of inter-individual CVs	90% inter-individual CVs
Control all urine	0.941	2.476
Male urine	0.890	2.273
Female urine	0.891	2.296
Aged 20-30	0.794	1.997
Aged 30-40	0.778	1.879
Aged 40-50	0.868	2.043
Aged 50-60	0.827	2.104
Aged >60	0.872	2.062
BCa all urine	0.833	2.071
BCa female urine	0.725	1.706
BCa male urine	0.785	1.700

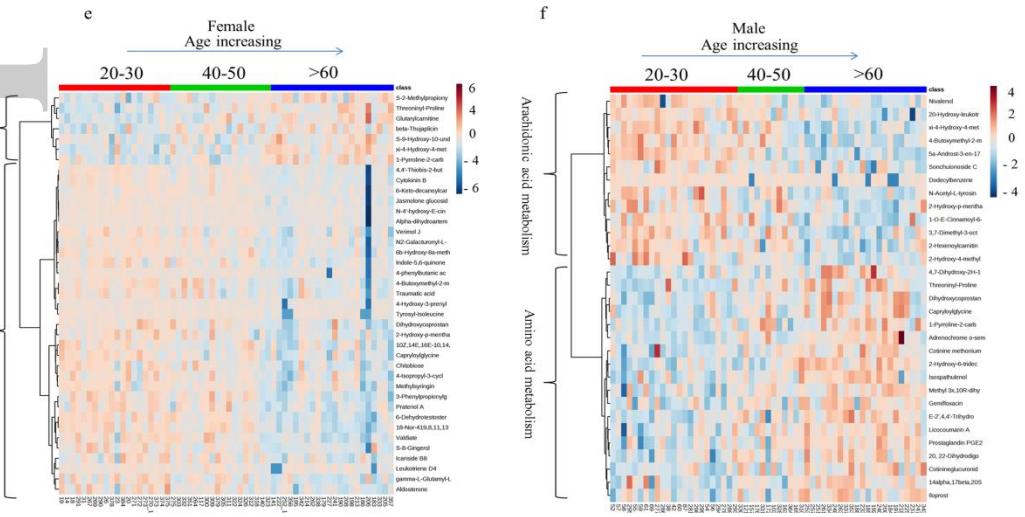
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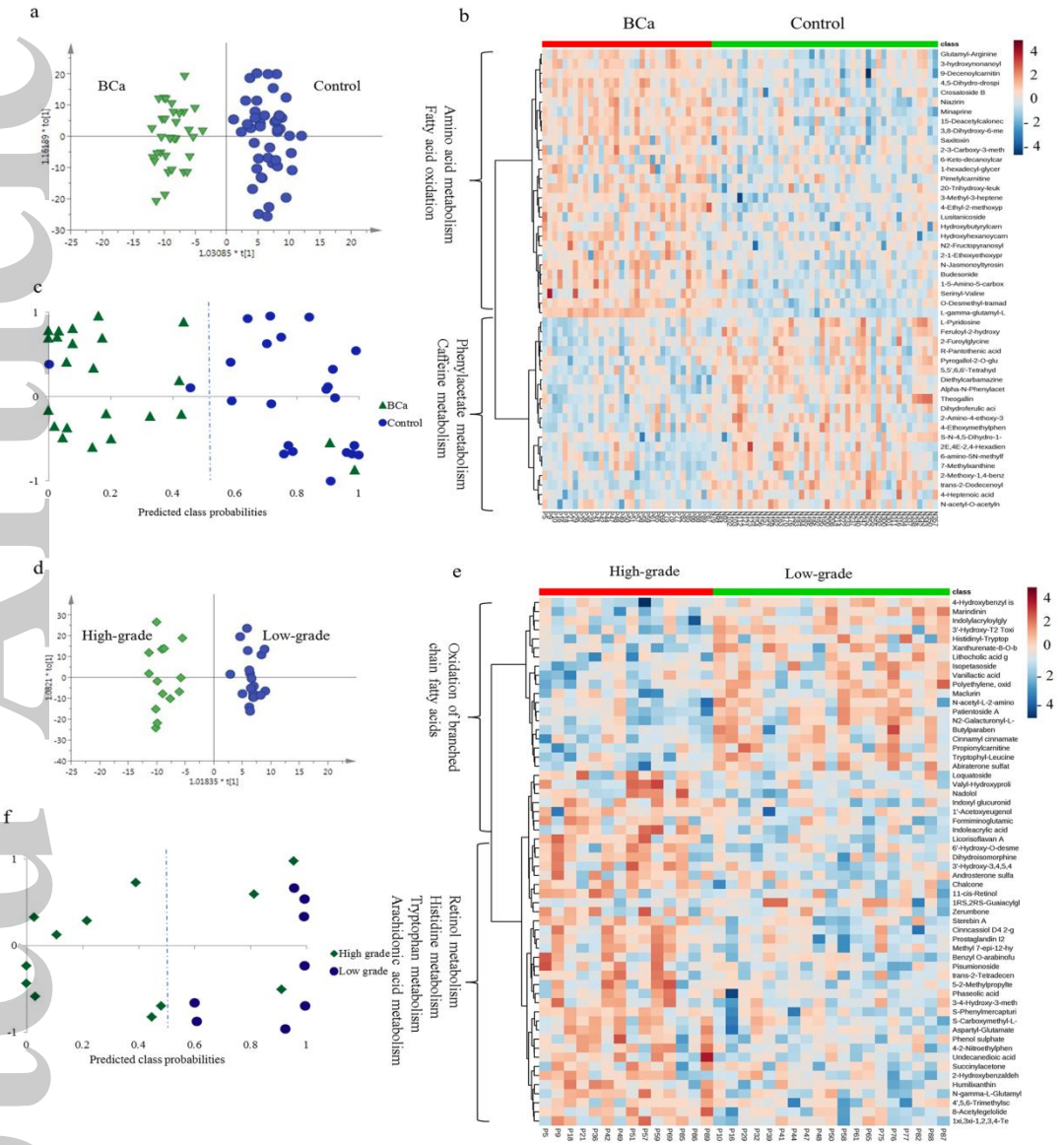






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